

On the Euler–Lagrange Equation for Planar Systems of Rigid Bodies or Lumped Masses

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The Euler–Lagrange equation is frequently used to develop the governing dynamic equilibrium expressions for rigid-body or lumped-mass systems. In many cases, however, the rectangular coordinates are constrained, necessitating either the use of Lagrange multipliers or the introduction of generalized coordinates that are consistent with the kinematic constraints. For such cases, evaluating the derivatives needed to obtain the governing equations can become a very laborious process. Motivated by several relevant problems related to rigid-body structures under seismic motions, this paper focuses on extending the elegant equations of motion developed by Greenwood in the 1970s, for the special case of planar systems of rigid bodies, to include rigid-body rotations and accelerating reference frames. The derived form of the Euler–Lagrange equation is then demonstrated with two examples: the double pendulum and a rocking object on a double rolling isolation system. The work herein uses an approach that is used by many analysts to derive governing equations for planar systems in translating reference frames (in particular, ground motions), but effectively precalculates some of the important stages of the analysis. It is hoped that beyond re-emphasizing the work by Greenwood, the specific form developed herein may help researchers save a significant amount of time, reduce the potential for errors in the formulation of the equations of motion for dynamical systems, and help introduce more researchers to the Euler–Lagrange equation. [DOI: 10.1115/1.4044145]

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1 Motivation and Background

Rigid-body systems are ubiquitous as subjects of study in physics and engineering. This is because many mechanical systems can be approximated using assemblies of rigid bodies. Three examples are shown in Fig. 1: (a) a classical double pendulum with lumped masses under support motion; (b) an assembly of tensioned rigid rocking blocks under base motion (e.g., earthquake

motion); and (c) a link model exhibiting snap-through. The double pendulum is a canonical problem in the study of dynamical systems [1], and has also seen recent use in base isolation applications in structural engineering [2]. Much of the motivation for this work, however, comes from systems such as the rocking block assembly in part (b) (restricting our discussion to the smooth motion between impacts). In Fig. 1(b), only the translation (u_4 and v_4) and rotation θ_4 are shown for brevity, but each block contributes to the angular and translational momentum of the system. In such systems, the calculation of the necessary derivatives of the kinetic energy is often a tedious task. While this task has been greatly alleviated by symbolic calculation software, it remains a barrier to the derivation of the governing equations, and limits the insights available from analytical expressions of the governing equations. Rocking systems in particular have been proposed as base isolators to protect buildings during earthquakes [3], and result in unwieldy governing equations [4,5] when investigated analytically, even for the smooth dynamics between impact events for systems of only two or three blocks. It should be noted that while the number of degrees-of-freedom and the number of masses/rigid bodies are equal in both of the first two examples, the formulation herein works in the general case. One simple example of this is shown in Fig. 1(c) (see Ref. [6]), which is a linkage system with multiple lumped masses (masses of the slider, m_s , and joint, m_c) and rigid bodies (masses of the two beams, m_b), but only one degree-of-freedom, θ .

The Euler–Lagrange equation [7] has become a standard tool in the study of vibrations. A detailed discussion of the equation with both holonomic and nonholonomic constraints was published by Greenwood [8,9]. An alternative formulation known as Kane’s method [10] can also be used to handle holonomic and nonholonomic constraints (and can be recovered from the Euler–Lagrange equation in the holonomic case [11]). This work provides an informative reformulation of the governing equations of motion that the authors hope will simplify calculations for experts, and also make the study of nonlinear dynamics more approachable for nonexperts. The equations obtained in Sec. 3 are a special case of those by Greenwood, and separately by Kalaba and Udwardia [12]. The version herein also directly includes reference frame translation (e.g., base motion) and rigid-body rotation. The formulation is demonstrated through two examples: (1) a double pendulum and (2) a system that includes both rocking and a double pendulum.

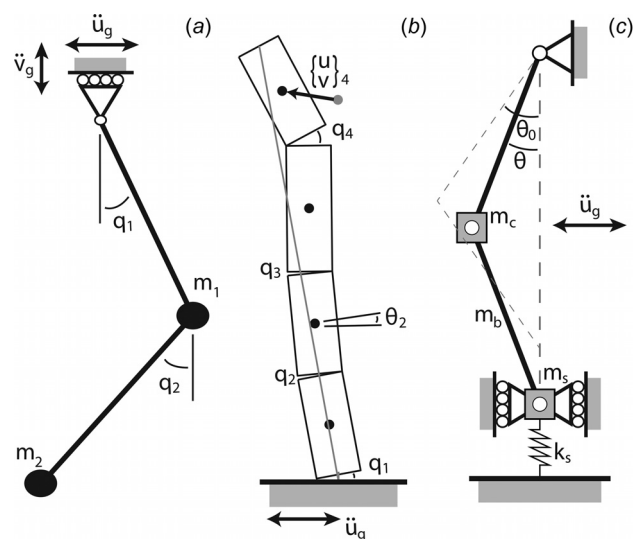


Fig. 1 Representative examples of planar rigid-body dynamical systems: (a) classical double pendulum, (b) rocking block assembly with tensioning cable from top block to ground, and (c) snap-through link model

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2 Kinematics

Assume that for a planar system of P rigid bodies or lumped masses, there exists a set of N independent generalized coordinates $\mathbf{q}$ such that

$$u_p = u_p(\mathbf{q}), \quad v_p = v_p(\mathbf{q}), \quad \theta_p = \theta_p(\mathbf{q}), \quad (p = 1, \dots, P) \quad (1)$$

where $(u, v, \theta)_p$ are the (potentially constrained) rectangular coordinates and rotation of the p th rigid body, and the vector $\mathbf{q}$ represents independent generalized coordinates that uniquely characterize the system configuration. The assumption is therefore made that independent generalized coordinates that are kinematically consistent with the system constraints can be found. Furthermore, note that the transformation equations do not contain time explicitly. In general, the horizontal acceleration of the p th rigid body is given by

$$\ddot{u}_p = \sum_{i=1}^N \frac{\partial u_p}{\partial q_i} \ddot{q}_i + \sum_{i=1}^N \sum_{j=1}^N \frac{\partial^2 u_p}{\partial q_i \partial q_j} \dot{q}_i \dot{q}_j \quad (2)$$

At this level of abstraction, the vertical and angular components present the same functional form and will therefore not be presented separately. The first partial derivatives in these expressions are also the components of the $P \times N$ gradient tensor $\mathbf{\nabla u}$, while the second derivatives are the components of the $P \times N \times N$ third-order tensor $\mathbf{\nabla \nabla u}$. Finally, the symmetry of the temporal (total time derivative) and spatial partial derivatives holds in this case, i.e.,

$$\frac{d}{dt} \frac{\partial u_p}{\partial q_n} = \frac{\partial}{\partial q_n} (\dot{u}_p) \quad (3)$$

3 The Euler–Lagrange Equation

The Euler–Lagrange equation of dynamic equilibrium, in terms of the Lagrangian $\mathcal{L} = \mathcal{T} - \mathcal{V}$, where $\mathcal{T}$ is the total kinetic energy and $\mathcal{V}$ is the potential energy, is given by [8]

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_n} \right) - \frac{\partial \mathcal{L}}{\partial q_n} = 0, \quad (n = 1, \dots, N) \quad (4)$$

In the typical case where the potential energy term is independent of the generalized velocity, this can be reduced to

$$\frac{d}{dt} \left(\frac{\partial \mathcal{T}}{\partial \dot{q}_n} \right) - \frac{\partial \mathcal{T}}{\partial q_n} + \frac{\partial \mathcal{V}}{\partial q_n} = 0, \quad (n = 1, \dots, N) \quad (5)$$

In cases where the inertia can be lumped at nodes, the total kinetic energy $\mathcal{T}$ can be separated into translational $\mathcal{T}_t$ (which can further be divided into horizontal $\mathcal{T}_u$ and vertical $\mathcal{T}_v$) and rotational $\mathcal{T}_\theta$ components, i.e.,

$$\begin{aligned} \mathcal{T} &= \mathcal{T}_u + \mathcal{T}_v + \mathcal{T}_\theta \\ &= \sum_{p=1}^P \frac{1}{2} m_p (\dot{u}_p + \dot{u}_g)^2 + \sum_{p=1}^P \frac{1}{2} m_p (\dot{v}_p + \dot{v}_g)^2 + \sum_{p=1}^P \frac{1}{2} I_p \dot{\theta}_p^2 \end{aligned} \quad (6)$$

where u_g and v_g denote the horizontal and vertical reference frame translation. Note that the formulation does not capture rotating reference frames. Moreover, note that, by expressing the coordinates u_p and v_p relative to the translating frame, the coordinate transformations (Eq. (1)) do not contain time explicitly, yet the system is rheonomic. This is readily apparent if the inertial coordinates x_p and y_p were used instead, in which case the coordinate transformations would explicitly depend on time, e.g., $x_p \equiv x_p(\mathbf{q}, t) = u_g(t) + u_p(\mathbf{q})$ [8,13].

The three terms $\mathcal{T}_u$, $\mathcal{T}_v$, and $\mathcal{T}_\theta$ present the same form (the form of the rotational term can be recovered from the other two if the base motions are set to zero), and thus it will be sufficient to

analyze just one of them in order to uncover the functional form of the equilibrium equations. The gradient of $\mathcal{T}_u$ has components

$$\frac{\partial \mathcal{T}_u}{\partial q_n} = \sum_{p=1}^P m_p (\dot{u}_p + \dot{u}_g) \frac{\partial \dot{u}_p}{\partial q_n} \quad (7)$$

Likewise, the partial derivative of $\mathcal{T}_u$ with respect to $\dot{q}_n$ is given by

$$\frac{\partial \mathcal{T}_u}{\partial \dot{q}_n} = \sum_{p=1}^P m_p (\dot{u}_p + \dot{u}_g) \frac{\partial \dot{u}_p}{\partial \dot{q}_n} = \sum_{p=1}^P m_p (\dot{u}_p + \dot{u}_g) \frac{\partial u_p}{\partial q_n} \quad (8)$$

where the distinction between the two versions is that the time derivatives in numerator and denominator of the partial derivative cancel out. Taking the total derivative of Eq. (8) with respect to time, and utilizing the symmetry of the time and spatial derivatives (shown in Eq. (3)) gives

$$\frac{d}{dt} \left(\frac{\partial \mathcal{T}_u}{\partial \dot{q}_n} \right) = \sum_{p=1}^P m_p (\ddot{u}_p + \ddot{u}_g) \frac{\partial u_p}{\partial q_n} + \sum_{p=1}^P m_p (\dot{u}_p + \dot{u}_g) \frac{\partial \dot{u}_p}{\partial q_n} \quad (9)$$

Finally, noting that second term in Eq. (9) cancels out Eq. (7) (also noted by Greenwood [8]), and using Eq. (2) to eliminate $\ddot{u}_p$, the kinetic energy components of the Euler–Lagrange equation reduce to

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial \mathcal{T}_u}{\partial \dot{q}_n} \right) - \frac{\partial \mathcal{T}_u}{\partial q_n} &= \sum_{p=1}^P m_p (\ddot{u}_p + \ddot{u}_g) \frac{\partial u_p}{\partial q_n} \\ &= \sum_{p=1}^P m_p \left(\sum_{i=1}^N \frac{\partial u_p}{\partial q_i} \ddot{q}_i + \sum_{i=1}^N \sum_{j=1}^N \frac{\partial^2 u_p}{\partial q_i \partial q_j} \dot{q}_i \dot{q}_j \right) \frac{\partial u_p}{\partial q_n} + \sum_{p=1}^P m_p \ddot{u}_g \frac{\partial u_p}{\partial q_n} \\ &= \sum_{i=1}^N \left(\sum_{p=1}^P m_p \frac{\partial u_p}{\partial q_i} \frac{\partial u_p}{\partial q_n} \right) \ddot{q}_i + \sum_{i=1}^N \sum_{j=1}^N \left(\sum_{p=1}^P m_p \frac{\partial^2 u_p}{\partial q_i \partial q_j} \frac{\partial u_p}{\partial q_n} \right) \dot{q}_i \dot{q}_j \\ &\quad + \left(\sum_{p=1}^P m_p \frac{\partial u_p}{\partial q_n} \right) \ddot{u}_g \\ &= \mathbf{M}_n^u \cdot \ddot{\mathbf{q}} + \dot{\mathbf{q}} \cdot \mathbf{G}_n^u \cdot \dot{\mathbf{q}} + \{\mathbf{F}_g^u\}_n \ddot{u}_g \end{aligned} \quad (10)$$

The vector $\mathbf{M}_n^u$ is the n th row (for the n th equation) of the square symmetric mass matrix $\mathbf{M}^u$, which has components

$$[\mathbf{M}^u]_{ni} = \sum_{p=1}^P m_p \frac{\partial u_p}{\partial q_i} \frac{\partial u_p}{\partial q_n} \quad (11)$$

the matrix $\mathbf{G}_n^u$ is a square symmetric matrix (for the horizontal displacement terms u) with the components

$$[\mathbf{G}_n^u]_{ij} = \sum_{p=1}^P m_p \frac{\partial^2 u_p}{\partial q_i \partial q_j} \frac{\partial u_p}{\partial q_n} \quad (12)$$

and the scalar $\{\mathbf{F}_g^u\}_n$ is the n th component of the ground motion-induced (or any other reference frame acceleration) force-pattern vector $\mathbf{F}_g^u$, which has components

$$\{\mathbf{F}_g^u\}_n = \sum_{p=1}^P m_p \frac{\partial u_p}{\partial q_n} \quad (13)$$

The results in Eq. (10) can also be written in vector form by “stacking” the $\mathbf{G}_n^u$ matrices for $n = 1, \dots, N$ into a third-order tensor. Care must be taken to ensure the pre- and postdot products are applied over the correct indices (i.e., the indices of the mixed second partial derivative) of $\mathbf{G}_n^u$ for each equation n . Making this final step, along with the inclusion of the vertical and rotational

components, results in the following vector expression of the Euler–Lagrange equation for systems with lumped inertia:

$$(\mathbf{M}^u + \mathbf{M}^v + \mathbf{M}^\theta) \cdot \ddot{\mathbf{q}} + \dot{\mathbf{q}} \cdot (\mathbf{G}^u + \mathbf{G}^v + \mathbf{G}^\theta) \cdot \dot{\mathbf{q}} + (\mathbf{F}_g^u) \ddot{u}_g + (\mathbf{F}_g^v) \ddot{v}_g + \nabla \mathcal{V} = \mathbf{0} \quad (14)$$

where the superscripts have been added to indicate the horizontal, vertical, and rotational degrees-of-freedom. For example, in the matrix $\mathbf{M}^v$ and third-order tensor $\mathbf{G}^v$, all instances of u in Eqs. (11), (12), and (13) are replaced with v . The equation does not include dissipation; however, damping terms are frequently appended (based on experience, or experimental identification) to the equations afterward. Note also that by replacing the angular momentum components with out-of-plane translation, the equations can be extended to 3D lumped mass systems without rotational inertia.

Equation (14) makes some effects of nonlinearity more apparent than is available by direct inspection of the Euler–Lagrange Eq. (4). First, the $\mathbf{F}_g$ vectors are (potentially) dependent on the generalized coordinates (although not the generalized velocity), meaning that reference frame motion may lead to parametric forcing. The $\mathbf{M}$ matrix and $\mathbf{G}$ tensor are likewise (potentially) dependent on the generalized coordinates. Additionally, any system with a linear transformation between the rectangular coordinates and the chosen independent generalized coordinates will result in a constant mass matrix and a constant force-pattern vector, and the $\mathbf{G}$ tensor will vanish (nonlinear restoring forces may still appear in the gradient of the potential). In the special case where the rectangular coordinates are equal to the independent general coordinates selected by the analyst (if this is possible without violating constraints), the mass matrix becomes the classical diagonal lumped-inertia matrix and the force-pattern vectors become constant vectors containing element masses as is expected for traditional undamped multidegree-of-freedom spring–mass systems. The second term in Eq. (14) also provides some helpful insights, as it shows that the generalized velocities must appear in quadratic forms only (linear velocity terms would be suspect as they dissipate energy). A final useful sanity check is that the equations are linear in terms of the reference frame accelerations.

For clarity, Eq. (14) can alternatively be expressed in summation notation as follows:

$$\begin{aligned} & \sum_{i=1}^N \sum_{p=1}^P \left(m_p \frac{\partial u_p}{\partial q_n} \frac{\partial u_p}{\partial q_i} + m_p \frac{\partial v_p}{\partial q_n} \frac{\partial v_p}{\partial q_i} + I_p \frac{\partial \theta_p}{\partial q_n} \frac{\partial \theta_p}{\partial q_i} \right) \ddot{q}_i \\ & + \sum_{i=1}^N \sum_{j=1}^N \sum_{p=1}^P \left(m_p \frac{\partial u_p}{\partial q_n} \frac{\partial^2 u_p}{\partial q_i \partial q_j} + m_p \frac{\partial v_p}{\partial q_n} \frac{\partial^2 v_p}{\partial q_i \partial q_j} + I_p \frac{\partial \theta_p}{\partial q_n} \frac{\partial^2 \theta_p}{\partial q_i \partial q_j} \right) \dot{q}_i \dot{q}_j \\ & + \left(\sum_{p=1}^P m_p \frac{\partial u_p}{\partial q_n} \right) \ddot{u}_g + \left(\sum_{p=1}^P m_p \frac{\partial v_p}{\partial q_n} \right) \ddot{v}_g + \frac{\partial \mathcal{V}}{\partial q_n} = 0, \quad (n = 1, \dots, N) \end{aligned} \quad (15)$$

An alternative derivation, based on a Lagrange multiplier formulation, is provided in the Appendix.

4 A Simple Demonstration

Consider the motion of a single particle ($P = 1$) of mass m constrained to move along the path $y = h(x)$ under the acceleration of gravity. Letting $u = q$, then $v = h(q)$ and the potential energy is given by $\mathcal{V} = mgv = mgh(q)$ where g is acceleration due to gravity. The objective is to write the equation of motion. Application of Newton’s second law is made more difficult as the particle is constrained and the reaction forces associated with the constraint condition need to be determined in the process. Using the method detailed herein, Eq. (15) results in

$$m \left[\left(\frac{\partial u}{\partial q} \right)^2 + \left(\frac{\partial v}{\partial q} \right)^2 \right] \ddot{q} + m \left(\frac{\partial u}{\partial q} \frac{\partial^2 u}{\partial q^2} + \frac{\partial v}{\partial q} \frac{\partial^2 v}{\partial q^2} \right) \dot{q}^2 + \frac{\partial \mathcal{V}}{\partial q} = 0 \quad (16)$$

Evaluating the four constraint derivatives [$\partial u / \partial q = 1$, $\partial^2 u / \partial q^2 = 0$, $\partial v / \partial q = h'(q)$, $\partial^2 v / \partial q^2 = h''(q)$], as well as the gradient of the potential energy [$\partial \mathcal{V} / \partial q = mgh'(q)$], we recover the equation of motion

$$m\{1 + [h'(q)]^2\} \ddot{q} + mh'(q)h''(q)\dot{q}^2 + mgh'(q) = 0 \quad (17)$$

Alternatively, the classical approach would be to use Lagrange’s equation directly to derive the equation of motion. Doing so necessitates the kinetic energy to be formulated

$$\begin{aligned} \mathcal{T} &= \frac{1}{2} m (\dot{u}^2 + \dot{v}^2) \\ &= \frac{1}{2} m \left[\left(\frac{\partial u}{\partial q} \dot{q} \right)^2 + \left(\frac{\partial v}{\partial q} \dot{q} \right)^2 \right] \\ &= \frac{1}{2} m \{ 1 + [h'(q)]^2 \} \dot{q}^2 \end{aligned} \quad (18)$$

which requires the evaluation of the derivatives $\partial u / \partial q$ and $\partial v / \partial q$. This direct approach (Eq. (5)) also requires four additional derivatives [$\partial \mathcal{T} / \partial \dot{q} = m\{1 + [h'(q)]^2\} \dot{q}$, $\partial^2 \mathcal{T} / (\partial \dot{q} \partial \dot{q}) = m\{1 + [h'(q)]^2\}$, $\partial^2 \mathcal{T} / (\partial q \partial \dot{q}) = 2mh'(q)h''(q)\dot{q}$, $\partial \mathcal{T} / \partial q = mh'(q)h''(q)\dot{q}^2$], where the two second derivatives arise from the total time derivative. The resulting equation is

$$m\{1 + [h'(q)]^2\} \ddot{q} + 2mh'(q)h''(q)\dot{q}^2 - mh'(q)h''(q)\dot{q}^2 + mgh'(q) = 0 \quad (19)$$

which, after combining the second and third terms, is identical to the result achieved by using the proposed form of the Euler–Lagrange equation (Eq. (15)).

4.1 Discussion. The formulation outlined here, while yielding an identical result as Lagrange’s equation, has a few attractive properties. First, unlike Lagrange’s equation, there is no need to determine the kinetic energy function. Second, the proposed formulation requires fewer derivatives to be taken. For this example, only four simple derivatives were required (not including the gradient of the potential energy) with the proposed formulation, while application of Lagrange’s equation required six derivatives that are more involved. Third, Lagrange’s equation produced additional terms that need to be combined (the terms also partially cancel), while the proposed formulation did not. The combination of these terms has already been handled in the development of the final form. Finally, there is no need to introduce the excess Lagrange multipliers (see Appendix), increasing the number of unknowns. This simple demonstration has shown some of the advantages of the proposed formulation. In Secs. 5 and 6, the formulation will be applied to two, more complex, problems.

5 Example 1: The Double Pendulum

The governing equations of the double pendulum are relatively easy to develop directly using the Euler–Lagrange equation. They are used here merely to demonstrate some of the calculations in Eq. (14) using this well-known example. The relationship between the rectangular and generalized coordinates (the two angles) in Fig. 1(a) is given by

$$\begin{aligned} u_1 &= L_1 \sin q_1, & v_1 &= -L_1 \cos q_1 \\ u_2 &= L_1 \sin q_1 + L_2 \sin q_2, & v_2 &= -L_1 \cos q_1 - L_2 \cos q_2 \end{aligned}$$

The potential energy is given by $\mathcal{V} = m_1 g v_1 + m_2 g v_2$. The definition in Eq. (11) yields the following mass matrix:

$$\mathbf{M}^{\text{tot}} = \mathbf{M}^u + \mathbf{M}^v = \begin{bmatrix} (m_1 + m_2)L_1^2 & m_2 L_1 L_2 \cos(q_1 - q_2) \\ m_2 L_1 L_2 \cos(q_1 - q_2) & m_2 L_2^2 \end{bmatrix} \quad (20)$$

where, for example, the first component is given by

$$\begin{aligned} [\mathbf{M}^{\text{tot}}]_{11} &= [\mathbf{M}^u]_{11} + [\mathbf{M}^v]_{11} \\ &= \left(m_1 \frac{\partial u_1}{\partial q_1} \frac{\partial u_1}{\partial q_1} + m_2 \frac{\partial u_2}{\partial q_1} \frac{\partial u_2}{\partial q_1} \right) \\ &\quad + \left(m_1 \frac{\partial v_1}{\partial q_1} \frac{\partial v_1}{\partial q_1} + m_2 \frac{\partial v_2}{\partial q_1} \frac{\partial v_2}{\partial q_1} \right) \\ &= (m_1 + m_2)L_1^2 \end{aligned} \quad (21)$$

Likewise, Eq. (12) yields

$$\begin{aligned} \mathbf{G}_1^{\text{tot}} &= \mathbf{G}_1^u + \mathbf{G}_1^v = \begin{bmatrix} 0 & 0 \\ 0 & m_2 L_1 L_2 \sin(q_1 - q_2) \end{bmatrix}, \\ \mathbf{G}_2^{\text{tot}} &= \mathbf{G}_2^u + \mathbf{G}_2^v = \begin{bmatrix} -m_2 L_1 L_2 \sin(q_1 - q_2) & 0 \\ 0 & 0 \end{bmatrix} \end{aligned} \quad (22)$$

where, for example, the off-diagonal terms (the two off-diagonal terms are identical due to symmetry) for equation $n = 1$ are given by

$$\begin{aligned} [\mathbf{G}_1^{\text{tot}}]_{12} &= [\mathbf{G}_1^u]_{12} + [\mathbf{G}_1^v]_{12} \\ &= \left(m_1 \frac{\partial^2 u_1}{\partial q_1 \partial q_2} \frac{\partial u_1}{\partial q_1} + m_2 \frac{\partial^2 u_2}{\partial q_1 \partial q_2} \frac{\partial u_2}{\partial q_1} \right) \\ &\quad + \left(m_1 \frac{\partial^2 v_1}{\partial q_1 \partial q_2} \frac{\partial v_1}{\partial q_1} + m_2 \frac{\partial^2 v_2}{\partial q_1 \partial q_2} \frac{\partial v_2}{\partial q_1} \right) = 0 \end{aligned} \quad (23)$$

Finally, Eq. (13) yields

$$\mathbf{F}_g^u = \left\{ \begin{matrix} (m_1 + m_2)L_1 \cos q_1 \\ m_2 L_2 \cos q_2 \end{matrix} \right\}, \quad \mathbf{F}_g^v = \left\{ \begin{matrix} (m_1 + m_2)L_1 \sin q_1 \\ m_2 L_2 \sin q_2 \end{matrix} \right\} \quad (24)$$

where $\mathbf{F}_g^v$ has the same form as the gradient of the gravitational potential energy, which will always be the case for problems with a constant gravity field. Combining these expressions using Eq. (14), along with the restoring force (i.e., the gradient of the potential energy), yields the equations of motion given by

$$\begin{aligned} (m_1 + m_2)L_1^2 \ddot{q}_1 + m_2 L_1 L_2 \cos(q_1 - q_2) \ddot{q}_2 \\ + m_2 L_1 L_2 \sin(q_1 - q_2) \dot{q}_2^2 + (m_1 + m_2)L_1 \cos q_1 \ddot{u}_g \\ + (m_1 + m_2)L_1 \sin q_1 (\ddot{v}_g + g) = 0 \end{aligned} \quad (25a)$$

$$\begin{aligned} m_2 L_1 L_2 \cos(q_1 - q_2) \ddot{q}_1 + m_2 L_2^2 \ddot{q}_2 - m_2 L_1 L_2 \sin(q_1 - q_2) \dot{q}_1^2 \\ + m_2 L_2 \cos q_2 \ddot{u}_g + m_2 L_2 \sin q_2 (\ddot{v}_g + g) = 0 \end{aligned} \quad (25b)$$

6 Example 2: Rocking Object on Double Rolling Isolation System

Consider a freestanding rigid block isolated by a double rolling isolation system (RIS) [2] from the horizontal and vertical ground

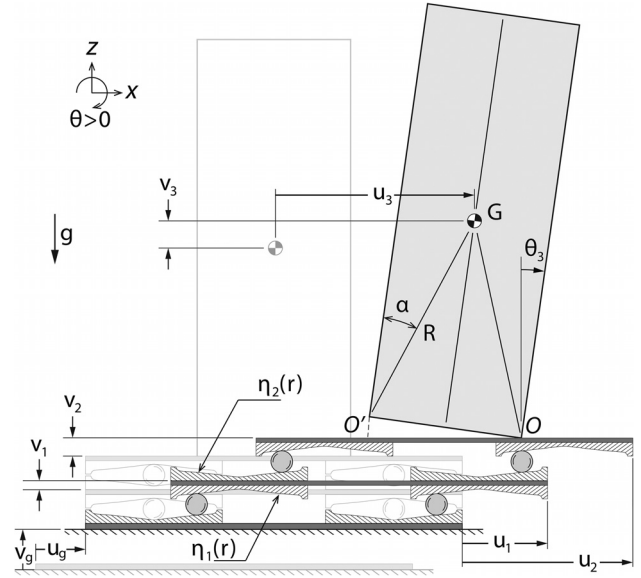


Fig. 2 Freestanding rigid block on a double rolling isolation system: geometry and displaced configuration

motions, $u_g(t)$ and $v_g(t)$, as shown in Fig. 2. The middle and top platforms of the RIS will displace relative to its base and middle platform, respectively, via the rolling of balls in counter-facing surfaces. The position of the RIS is measured by lateral displacements $u_1(t)$ and $u_2(t)$ of the middle and top platforms relative to the ground, as well as the vertical displacements $v_1(t)$ and $v_2(t)$. The vertical displacements are governed by the holonomic constraints [14]

$$v_1 = 2\eta_1(u_1/2), \quad v_2 = v_1 + 2\eta_2((u_2 - u_1)/2) \quad (26)$$

where $\eta_i(\cdot)$ is the lower and upper rolling surface radial profiles in the i th isolation layer.

The significant geometric property of the block is the location of its center of gravity G which, as shown in Fig. 2, is a distance R from a corner point at an angle α with the vertical. When set to rocking, the block will oscillate about the center of rotation O . The rotation of the block is measured by the angle $\theta_3(t)$ from the vertical. As is commonly done [15], it is assumed that the coefficient of friction is sufficiently large to prevent sliding between the block and the platform. Furthermore, for the purpose of demonstrating the method in this paper, the equations of motion for the system are derived when it is set to rocking only; modeling the impact, when the angle of rotation is reversed, is commonly treated using conservation of momentum [16,17], which is outside the scope of this paper.

The significant inertial property of the block are its mass m_3 and its moment of inertia I_3 about its center of mass, which for a rectangular block is $I_3 = m_3 R^2/3$. The middle platform and isolated base of the double RIS have masses m_1 and m_2 , respectively. The potential energy of the system is given by

$$\mathcal{V}(v_1, v_2, v_3) = m_1 g v_1 + m_2 g v_2 + m_3 g v_3 \quad (27)$$

The relationships between the rectangular and generalized coordinates are given by

$$u_1 = q_1, \quad u_2 = q_1 + q_2, \quad u_3 = q_1 + q_2 + R[\sin(\alpha) - \sin(\alpha - q_3)]$$

$$v_1 = 2\eta_1(q_1/2), \quad v_2 = 2\eta_1(q_1/2) + 2\eta_2(q_2/2),$$

$$v_3 = 2\eta_1(q_1/2) + 2\eta_2(q_2/2) + R \cos(\alpha - q_3)$$

and $\theta_3 = q_3$. Application of Eq. (14) immediately yields the equations of motion. The mass matrix is found to be

$$\mathbf{M}^{\text{tot}} = \begin{bmatrix} (m_1 + m_2 + m_3)(1 + \eta_1'^2) & (m_2 + m_3)(1 + \eta_1'\eta_2') & m_3R(\cos \vartheta + \eta_1' \sin \vartheta) \\ (m_2 + m_3)(1 + \eta_1'\eta_2') & (m_2 + m_3)(1 + \eta_2'^2) & m_3R(\cos \vartheta + \eta_2' \sin \vartheta) \\ m_3R(\cos \vartheta + \eta_1' \sin \vartheta) & m_3R(\cos \vartheta + \eta_2' \sin \vartheta) & I_3 + m_3R^2 \end{bmatrix} \quad (28)$$

where $\eta_1' \equiv \eta_1'(q_1/2)$, $\eta_2' \equiv \eta_2'(q_2/2)$, and $\vartheta \equiv \alpha - q_3$. The three matrices constituting the $\mathbf{G}$ tensor are found to be

$$\mathbf{G}_1^{\text{tot}} = \begin{bmatrix} \frac{1}{2}(m_1 + m_2 + m_3)\eta_1''\eta_1' & 0 & 0 \\ 0 & \frac{1}{2}(m_2 + m_3)\eta_2''\eta_1' & 0 \\ 0 & 0 & -m_3\eta_1'R \cos \vartheta \end{bmatrix} \quad (29a)$$

$$\mathbf{G}_2^{\text{tot}} = \begin{bmatrix} \frac{1}{2}(m_2 + m_3)\eta_1''\eta_2' & 0 & 0 \\ 0 & \frac{1}{2}(m_2 + m_3)\eta_2''\eta_2' & 0 \\ 0 & 0 & -m_3\eta_2'R \cos \vartheta \end{bmatrix} \quad (29b)$$

$$\mathbf{G}_3^{\text{tot}} = \begin{bmatrix} \frac{1}{2}m_3\eta_1''R \sin \vartheta & 0 & 0 \\ 0 & \frac{1}{2}m_3\eta_2''R \sin \vartheta & 0 \\ 0 & 0 & -m_3R^2 \cos \vartheta \sin \vartheta \end{bmatrix} \quad (29c)$$

where $\eta_1'' \equiv \eta_1''(q_1/2)$ and $\eta_2'' \equiv \eta_2''(q_2/2)$. Equation (13) gives

$$\mathbf{F}_g^u = \begin{Bmatrix} m_1 + m_2 + m_3 \\ m_2 + m_3 \\ m_3R \cos \vartheta \end{Bmatrix}, \quad \mathbf{F}_g^v = \begin{Bmatrix} (m_1 + m_2 + m_3)\eta_1' \\ (m_2 + m_3)\eta_2' \\ m_3R \sin \vartheta \end{Bmatrix} \quad (30)$$

Finally, the gradient of the potential energy is determined to be

$$\nabla \mathcal{V} = \begin{Bmatrix} (m_1 + m_2 + m_3)g\eta_1' \\ (m_2 + m_3)g\eta_2' \\ m_3gR \sin \vartheta \end{Bmatrix} \quad (31)$$

Combining these expressions using Eq. (14) yields the equations of motion

$$\begin{aligned} & (m_1 + m_2 + m_3)(1 + \eta_1'^2)\ddot{q}_1 + (m_2 + m_3)(1 + \eta_1'\eta_2')\ddot{q}_2 + m_3R[\cos(\alpha - q_3) + \eta_1' \sin(\alpha - q_3)]\ddot{q}_3 \\ & + \frac{1}{2}(m_1 + m_2 + m_3)h''_1\eta_1'\dot{q}_1^2 + \frac{1}{2}(m_2 + m_3)\eta_2''\eta_1'\dot{q}_2^2 - m_3R \cos(\alpha - q_3)\eta_1'\dot{q}_3^2 \\ & + (m_1 + m_2 + m_3)g\eta_1' = -(m_1 + m_2 + m_3)\ddot{u}_g - (m_1 + m_2 + m_3)\eta_1'\ddot{v}_g \end{aligned} \quad (32a)$$

$$\begin{aligned} & (m_2 + m_3)(1 + \eta_1'\eta_2')\ddot{q}_1 + (m_2 + m_3)(1 + \eta_2'^2)\ddot{q}_2 + m_3R[\cos(\alpha - q_3) + \eta_2' \sin(\alpha - q_3)]\ddot{q}_3 \\ & + \frac{1}{2}(m_2 + m_3)h''_1\eta_2'\dot{q}_1^2 + \frac{1}{2}(m_2 + m_3)\eta_2''\eta_2'\dot{q}_2^2 - m_3R \cos(\alpha - q_3)\eta_2'\dot{q}_3^2 \\ & + (m_2 + m_3)g\eta_2' = -(m_2 + m_3)\ddot{u}_g - (m_2 + m_3)\eta_2'\ddot{v}_g \end{aligned} \quad (32b)$$

$$\begin{aligned} & m_3R[\cos(\alpha - q_3) + \eta_1' \sin(\alpha - q_3)]\ddot{q}_1 + m_3R[\cos(\alpha - q_3) + \eta_2' \sin(\alpha - q_3)]\ddot{q}_2 + (I_3 + m_3R^2)\ddot{q}_3 \\ & + \frac{1}{2}m_3h''_1R \sin(\alpha - q_3)\dot{q}_1^2 + \frac{1}{2}m_3\eta_2''R \sin(\alpha - q_3)\dot{q}_2^2 - m_3R^2 \cos(\alpha - q_3)\sin(\alpha - q_3)\dot{q}_3^2 \\ & + m_3gR \sin(\alpha - q_3) = -m_3R \cos(\alpha - q_3)\ddot{u}_g - m_3R \sin(\alpha - q_3)\ddot{v}_g \end{aligned} \quad (32c)$$

Note that Eq. (32) was derived for $\theta > 0$ (rotation about point O), but a similar procedure can be followed to find the equations of motion for $\theta < 0$ (rotation about point O') [18].

7 Conclusions

This paper presents a special version of the Euler–Lagrange equation for cases where the inertia can be lumped at nodes. The derivation is focused on translation and rotation of planar rigid-body systems in an accelerating reference frame; however, it could be readily applied to lumped-mass systems in 3D (i.e., masses without rotational inertia). The calculations result in a system of ordinary differential equations in the generalized coordinates that have many useful characteristics. Two examples for which the new approach can be used to obtain the governing equations explicitly are shown.

It is hoped that at the very least this work will help researchers save time in the derivation of governing equations in the study of dynamical systems. In systems with complicated geometry, the calculation of the necessary derivatives of the kinetic energy is often a very involved and error prone process.

It is further hoped that these results can also find use in the engineering practice. While the principles of dynamic equilibrium and minimum potential energy are frequently covered in undergraduate curricula, Lagrangian mechanics is often left for graduate studies. The fact that the equations derived herein will look very familiar to practitioners may help broaden the appeal of nonlinear analysis.

Appendix: Lagrange Multiplier Approach

This section presents an alternative derivation of Eq. (15), namely through the use of Lagrange multipliers to enforce the coordinate transformations (Eq. (1)). If we consider the set $\{u, v, \theta, q\}$ to be generalized coordinates, then the coordinate transformations can be considered holonomic constraints in terms of these $(3P + N)$ generalized coordinates as follows:

$$f_p(u, q) = u_p - u_p(\mathbf{q}) = 0, \quad (p = 1, \dots, P) \quad (\text{A1a})$$

$$g_p(v, q) = v_p - v_p(\mathbf{q}) = 0, \quad (p = 1, \dots, P) \quad (\text{A1b})$$

$$h_p(\theta, q) = \theta_p - \theta_p(\mathbf{q}) = 0. \quad (p = 1, \dots, P) \quad (\text{A1c})$$

Lagrange multipliers λ , μ , and ν are used to enforce these constraints. Applying the holonomic form of Lagrange's equation results in the governing equations of motion: for coordinates u

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{u}_p} \right) - \frac{\partial \mathcal{L}}{\partial u_p} &= \sum_{k=1}^P \left(\frac{\partial f_k}{\partial u_p} \lambda_k + \frac{\partial g_k}{\partial u_p} \mu_k + \frac{\partial h_k}{\partial u_p} \nu_k \right) \\ \Rightarrow m_p(\ddot{u}_p + \ddot{u}_g) + \frac{\partial \mathcal{V}}{\partial u_p} &= \lambda_p, \quad (p = 1, \dots, P) \end{aligned} \quad (\text{A2})$$

for coordinates v

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{v}_p} \right) - \frac{\partial \mathcal{L}}{\partial v_p} &= \sum_{k=1}^P \left(\frac{\partial f_k}{\partial v_p} \lambda_k + \frac{\partial g_k}{\partial v_p} \mu_k + \frac{\partial h_k}{\partial v_p} \nu_k \right) \\ \Rightarrow m_p(\ddot{v}_p + \ddot{v}_g) + \frac{\partial \mathcal{V}}{\partial v_p} &= \mu_p, \quad (p = 1, \dots, P) \end{aligned} \quad (\text{A3})$$

for coordinates θ

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}_p} \right) - \frac{\partial \mathcal{L}}{\partial \theta_p} &= \sum_{k=1}^P \left(\frac{\partial f_k}{\partial \theta_p} \lambda_k + \frac{\partial g_k}{\partial \theta_p} \mu_k + \frac{\partial h_k}{\partial \theta_p} \nu_k \right) \\ \Rightarrow I_p \ddot{\theta}_p + \frac{\partial \mathcal{V}}{\partial \theta_p} &= \nu_p, \quad (p = 1, \dots, P) \end{aligned} \quad (\text{A4})$$

and for coordinates q

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_n} \right) - \frac{\partial \mathcal{L}}{\partial q_n} &= \sum_{p=1}^P \left(\frac{\partial f_p}{\partial q_n} \lambda_p + \frac{\partial g_p}{\partial q_n} \mu_p + \frac{\partial h_p}{\partial q_n} \nu_p \right) \\ \Rightarrow 0 &= \sum_{p=1}^P \left(-\frac{\partial u_p}{\partial q_n} \lambda_p - \frac{\partial v_p}{\partial q_n} \mu_p - \frac{\partial \theta_p}{\partial q_n} \nu_p \right), \quad (n = 1, \dots, N) \end{aligned} \quad (\text{A5})$$

$3P$ additional differential equations are needed since there are $6P + N$ variables ($u, v, \theta, q, \lambda, \mu, \nu$). Differentiating Eq. (A1) with respect to time twice, we obtain

$$\ddot{u}_p - \sum_{i=1}^N \sum_{j=1}^N \frac{\partial^2 u_p}{\partial q_i \partial q_j} \dot{q}_i \dot{q}_j - \sum_{i=1}^N \frac{\partial u_p}{\partial q_i} \ddot{q}_i = 0, \quad (p = 1, \dots, P) \quad (\text{A6a})$$

$$\ddot{v}_p - \sum_{i=1}^N \sum_{j=1}^N \frac{\partial^2 v_p}{\partial q_i \partial q_j} \dot{q}_i \dot{q}_j - \sum_{i=1}^N \frac{\partial v_p}{\partial q_i} \ddot{q}_i = 0, \quad (p = 1, \dots, P) \quad (\text{A6b})$$

$$\ddot{\theta}_p - \sum_{i=1}^N \sum_{j=1}^N \frac{\partial^2 \theta_p}{\partial q_i \partial q_j} \dot{q}_i \dot{q}_j - \sum_{i=1}^N \frac{\partial \theta_p}{\partial q_i} \ddot{q}_i = 0. \quad (p = 1, \dots, P) \quad (\text{A6c})$$

Using Eqs. (A2), (A3), and (A4) in Eq. (A5), one finds

$$\begin{aligned} \sum_{p=1}^P \left\{ -\frac{\partial u_p}{\partial q_n} \left[m_p(\ddot{u}_p + \ddot{u}_g) + \frac{\partial \mathcal{V}}{\partial u_p} \right] - \frac{\partial v_p}{\partial q_n} \left[m_p(\ddot{v}_p + \ddot{v}_g) + \frac{\partial \mathcal{V}}{\partial v_p} \right] \right. \\ \left. - \frac{\partial \theta_p}{\partial q_n} \left[I_p \ddot{\theta}_p + \frac{\partial \mathcal{V}}{\partial \theta_p} \right] \right\} = 0, \quad (n = 1, \dots, N) \end{aligned} \quad (\text{A7})$$

Multiplying through by -1 and noting that

$$\sum_{p=1}^P \left(\frac{\partial \mathcal{V}}{\partial u_p} \frac{\partial u_p}{\partial q_n} + \frac{\partial \mathcal{V}}{\partial v_p} \frac{\partial v_p}{\partial q_n} + \frac{\partial \mathcal{V}}{\partial \theta_p} \frac{\partial \theta_p}{\partial q_n} \right) \equiv \frac{\partial \mathcal{V}}{\partial q_n}, \quad (n = 1, \dots, N)$$

Equation (A7) can be rewritten

$$\begin{aligned} \sum_{p=1}^P \left[m_p \frac{\partial u_p}{\partial q_n} (\ddot{u}_p + \ddot{u}_g) + m_p \frac{\partial v_p}{\partial q_n} (\ddot{v}_p + \ddot{v}_g) + I_p \frac{\partial \theta_p}{\partial q_n} \ddot{\theta}_p \right] \\ + \frac{\partial \mathcal{V}}{\partial q_n} = 0, \quad (n = 1, \dots, N) \end{aligned} \quad (\text{A8})$$

Using Eq. (A6) in Eq. (A8), one finds

$$\begin{aligned} \sum_{p=1}^P \left\{ m_p \frac{\partial u_p}{\partial q_n} \left[\left(\sum_{i=1}^N \sum_{j=1}^N \frac{\partial^2 u_p}{\partial q_i \partial q_j} \dot{q}_i \dot{q}_j + \sum_{i=1}^N \frac{\partial u_p}{\partial q_i} \ddot{q}_i \right) + \ddot{u}_g \right] \right. \\ \left. + m_p \frac{\partial v_p}{\partial q_n} \left[\left(\sum_{i=1}^N \sum_{j=1}^N \frac{\partial^2 v_p}{\partial q_i \partial q_j} \dot{q}_i \dot{q}_j + \sum_{i=1}^N \frac{\partial v_p}{\partial q_i} \ddot{q}_i \right) + \ddot{v}_g \right] \right. \\ \left. + I_p \frac{\partial \theta_p}{\partial q_n} \left[\left(\sum_{i=1}^N \sum_{j=1}^N \frac{\partial^2 \theta_p}{\partial q_i \partial q_j} \dot{q}_i \dot{q}_j - \sum_{i=1}^N \frac{\partial \theta_p}{\partial q_i} \ddot{q}_i \right) \right] \right\} \\ + \frac{\partial \mathcal{V}}{\partial q_n} = 0, \quad (n = 1, \dots, N) \end{aligned} \quad (\text{A9})$$

and gathering terms, Eq. (15) is recovered.

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